

ELBOW, SK, Canada

WMO: 714500

Lat: 51.1256N

Lon: 106.5826W

Elev: 1952

StdP: 13.69

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-20.4	-31.5	1.0	-25.3	-26.0	1.4	-20.3	30.8	14.1	27.5	12.8	7.1	290	0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	88.3	64.8	84.3	63.6	80.5	62.1	69.0	81.3	66.7	78.9	64.7	76.3	11.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	99.6	74.5	62.5	91.0	71.4	60.4	84.3	69.4	34.4	81.1	32.4	78.7	30.8	76.1	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	23.6	20.9	-30.9	95.1	4.1	5.2	-33.9	98.8	-36.3	101.9	-38.7	104.8	-41.7	108.6
			-31.0	73.4	3.9	2.4	-33.8	75.1	-36.1	76.5	-38.3	77.8	-41.2	79.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	7.0	10.3	23.4	39.6	51.5	60.3	65.7	64.2	55.2	40.8	25.0	11.8	
	DBStd	23.58	15.62	14.34	14.55	10.03	8.22	6.18	5.47	6.33	8.14	9.04	12.30	14.59	
	HDD50	5961	1332	1111	824	329	81	4	0	1	40	304	751	1184	
	HDD65	10020	1797	1531	1289	762	422	164	57	92	307	749	1201	1649	
	CDD50	1601	0	0	0	17	129	313	487	440	195	20	0	0	
	CDD65	186	0	0	0	0	4	24	79	67	12	0	0	0	
	CDH74	2842	0	0	0	10	149	360	1003	1004	308	8	0	0	
	CDH80	915	0	0	0	2	28	89	331	370	95	0	0	0	
Wind	WSAvg	10.2	10.2	9.8	11.0	11.4	11.4	10.1	8.8	9.0	9.7	10.4	10.4	9.9	
Precipitation	PrecAvg	14.9	0.7	0.5	0.6	1.1	1.8	3.0	2.1	1.7	1.2	0.8	0.5	0.6	
	PrecMax	22.3	1.3	1.3	1.5	3.3	4.6	6.0	4.9	4.2	3.2	2.7	1.6	1.6	
	PrecMin	6.0	0.1	0.2	0.0	0.2	0.0	0.4	0.4	0.1	0.1	0.0	0.0	0.2	
	PrecStd	3.5	0.3	0.2	0.4	0.8	1.3	1.5	1.2	1.0	0.9	0.6	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	42.4	57.8	73.6	83.9	88.2	93.4	95.2	88.1	74.0	59.1	44.8	
		MCWB	36.7	36.6	45.9	53.1	57.6	63.4	69.5	66.9	61.5	55.7	47.2	38.1	
	2%	DB	37.5	37.1	50.8	67.5	78.1	81.9	87.6	88.8	82.4	66.2	49.8	39.6	
		MCWB	33.9	33.4	42.2	49.7	56.1	62.6	67.0	64.4	59.6	51.3	42.0	35.0	
	5%	DB	34.1	33.9	45.2	62.1	73.5	78.0	83.5	83.8	76.8	60.0	44.4	35.8	
		MCWB	31.4	31.5	39.0	47.1	53.8	60.2	65.9	63.2	57.7	48.4	38.7	32.8	
	10%	DB	29.2	30.7	40.2	56.7	68.8	74.2	79.5	79.4	71.3	55.0	39.9	31.6	
		MCWB	27.4	28.9	36.0	44.8	51.5	58.7	64.1	61.8	55.0	45.8	36.0	29.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	36.4	47.0	54.8	61.3	69.9	74.2	70.2	64.6	56.6	48.0	38.3
			MCDB	41.4	40.2	56.1	68.0	76.3	81.2	86.6	85.4	82.2	71.4	59.1	44.1
2%		WB	34.2	33.4	42.5	51.1	58.1	65.6	70.1	67.5	61.2	52.7	42.7	35.0	
		MCDB	37.2	36.0	50.0	64.3	73.2	76.7	81.5	81.1	77.2	63.6	49.7	38.8	
5%		WB	31.7	31.2	38.9	48.2	55.7	62.8	67.8	65.6	58.9	49.4	39.4	32.4	
		MCDB	34.5	33.3	44.9	60.7	70.4	73.0	79.4	78.7	73.8	57.7	44.3	35.3	
10%		WB	28.0	28.4	35.9	45.3	53.5	60.6	65.7	63.7	56.6	46.8	36.2	29.1	
		MCDB	29.9	30.1	39.5	55.0	66.9	71.3	76.8	75.9	69.1	54.5	39.6	31.1	
Mean Daily Temperature Range	5% DB	MDBR	17.9	18.2	17.9	21.8	24.6	21.2	23.3	24.9	24.3	20.3	16.1	16.3	
		MCDBR	19.8	19.5	22.3	32.0	32.6	28.4	30.5	33.8	35.4	29.7	23.0	19.4	
		MCWBR	17.0	16.9	15.0	16.5	14.9	12.9	13.8	13.5	15.6	16.7	15.3	15.3	
	5% WB	MCDBR	20.1	19.2	22.3	29.3	29.0	23.7	27.0	27.7	31.0	27.7	22.1	19.1	
		MCWBR	17.5	16.6	15.1	16.2	13.8	12.2	13.8	13.0	14.5	16.3	15.1	15.7	
Clear-Sky Solar Irradiance	taub		0.257	0.254	0.270	0.334	0.350	0.355	0.365	0.370	0.325	0.282	0.252	0.223	
	taud		2.304	2.363	2.488	2.369	2.372	2.409	2.391	2.356	2.460	2.559	2.493	2.410	
	Ebn at Noon		247	285	301	288	287	285	280	272	275	272	254	250	
	Edh at Noon		20	25	27	35	36	35	36	35	28	21	17	15	
All-Sky Solar Radiation	RadAvg		366	681	1118	1520	1860	1913	2064	1659	1217	731	380	269	
	RadStd		38	51	88	129	138	139	119	121	111	95	44	27	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.85	-2.27	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page